PROBLEM STATEMENTS

The goal of this analysis is to:

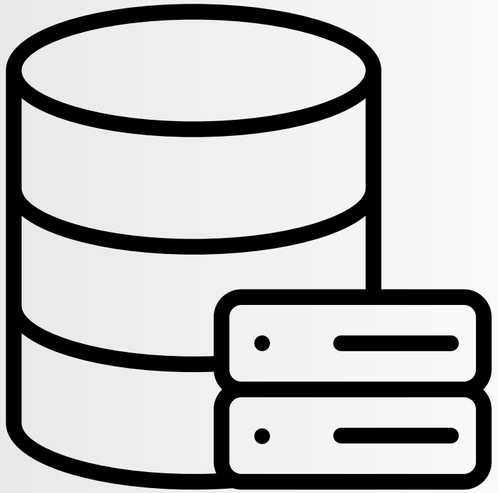
- Understand sales trends over time (monthly, weekly, hourly)
- Identify top customers and products
- Analyze country-wise contribution to revenue
- Detect patterns to improve sales performance and decision-making

Data Source

- **Dataset: E-commerce transaction data (CSV)**
- **Source: Database**
- **Key columns used:**

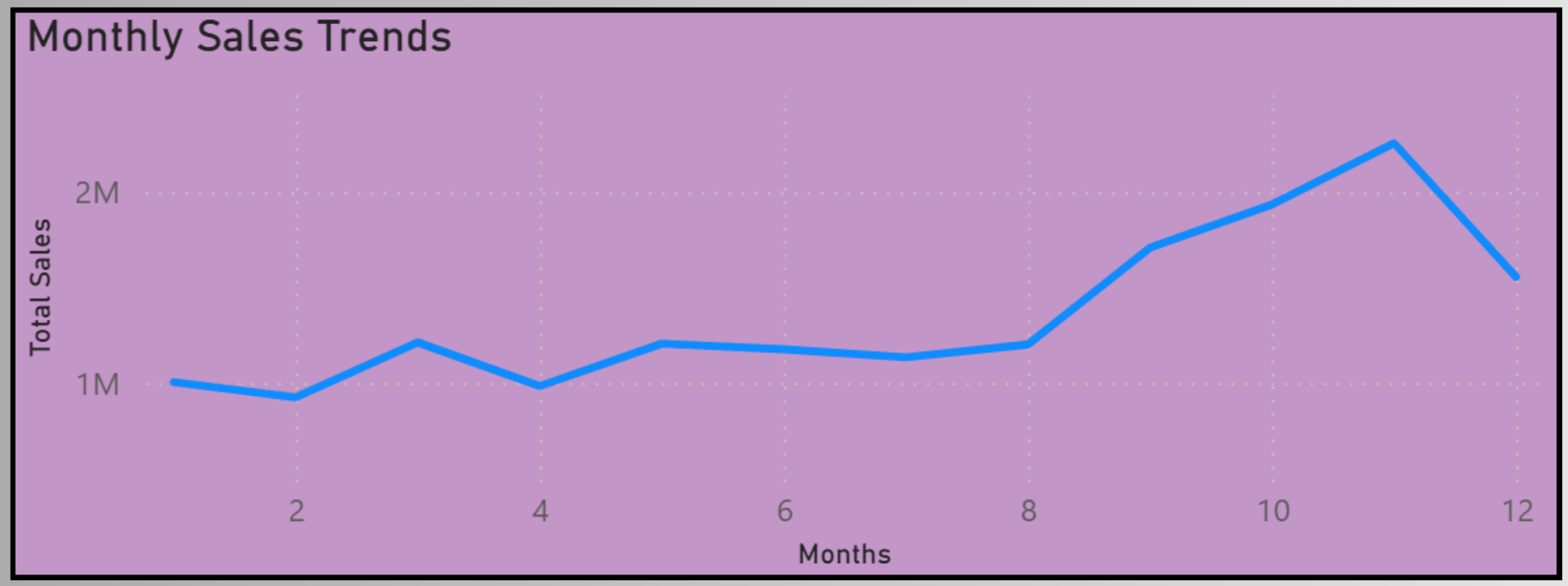
  1. Invoice
  2. StockCode
  3. Description
  4. Quantity
  5. InvoiceDate
  6. Price
  7. Customer ID
  8. Country

- **There are 1M rows and 8 Column in Dataset**

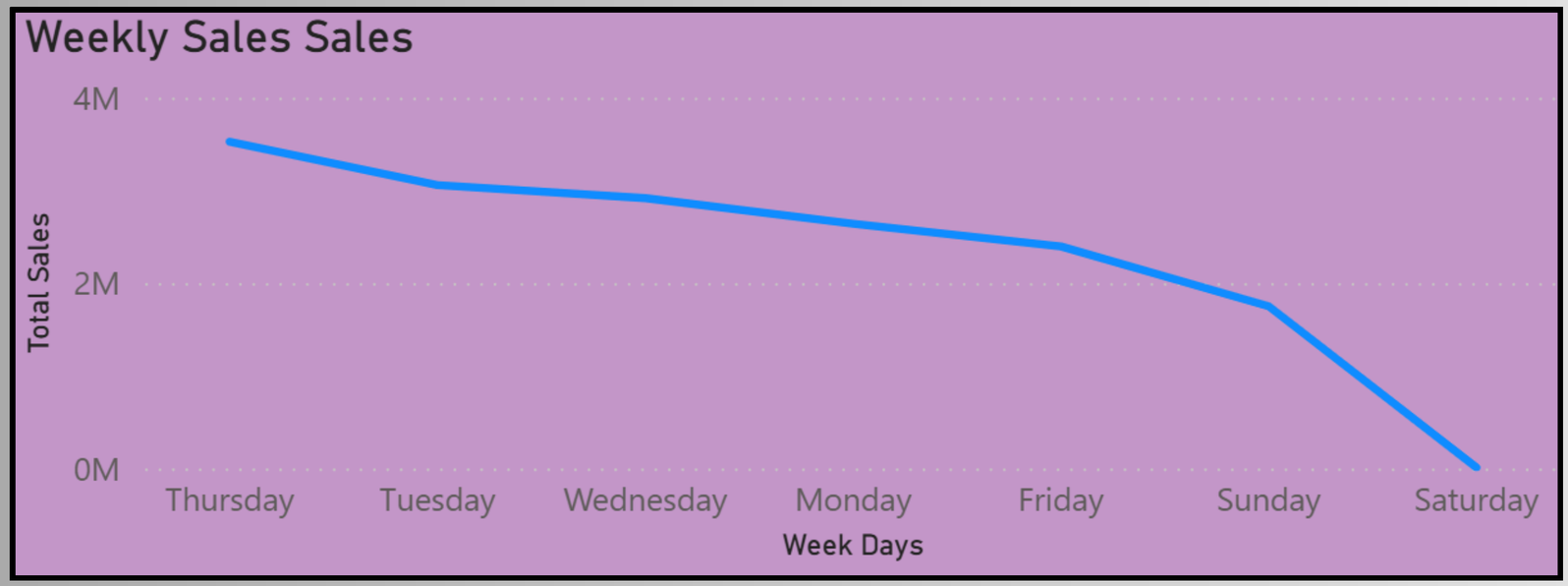


Analysis / Findings

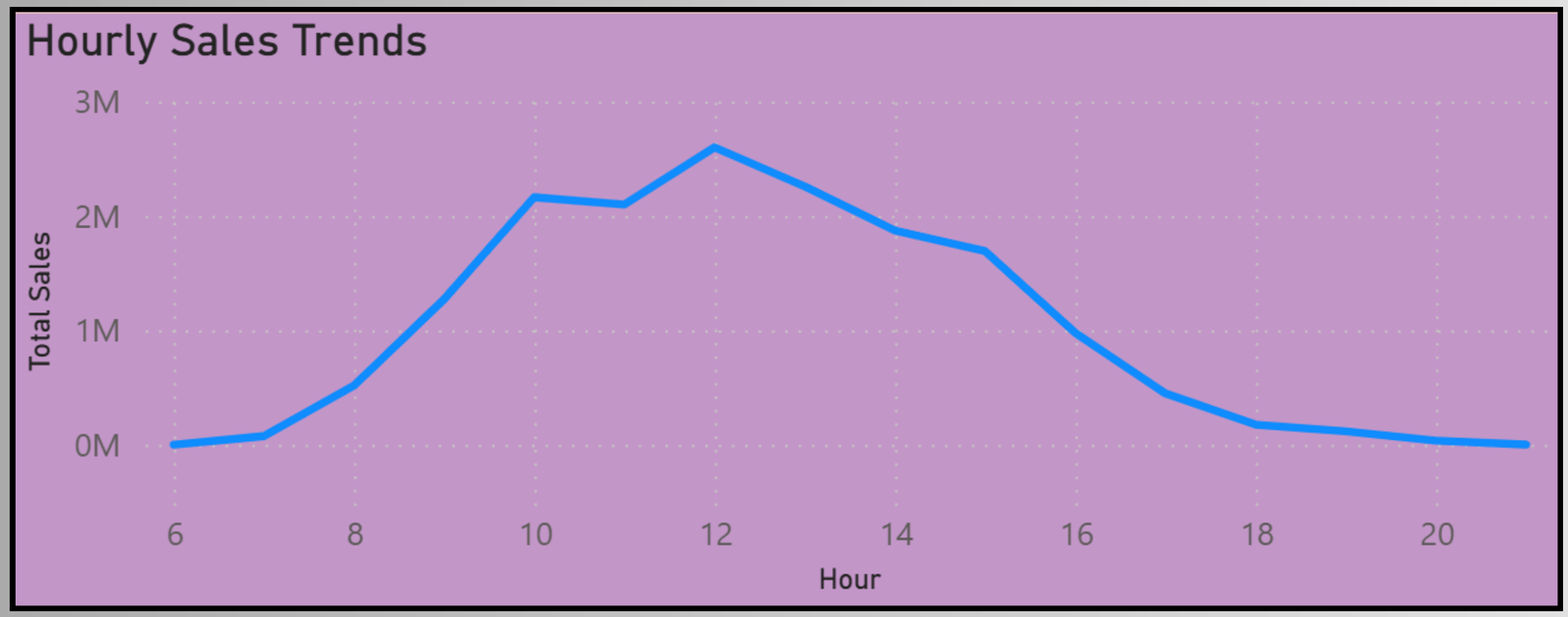
1. Sales Trends



- Monthly sales show growth towards end of the year, peaking around November Month
- 50% of Total Revenue Comes from Top 4 Months (Sep-Dec)

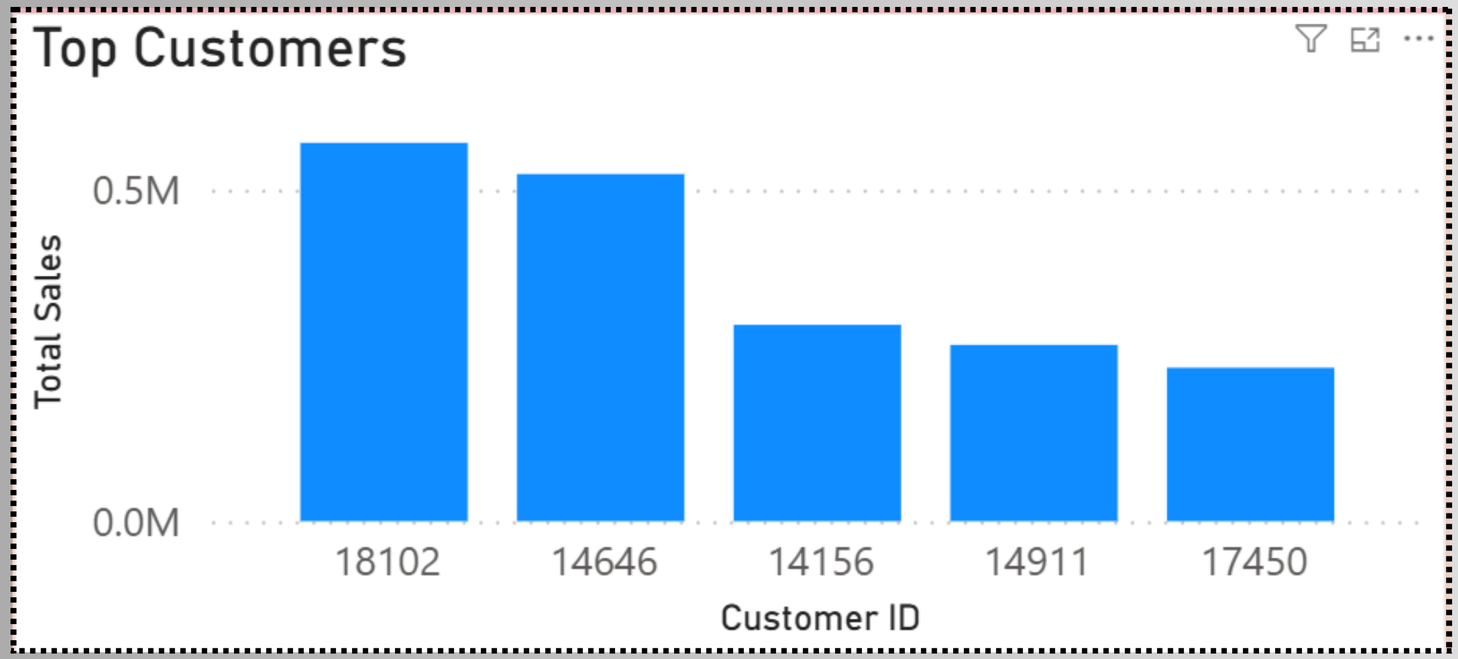


- Weekly sales show declining trend towards weekend, with lowest on Saturday and Highest on Thursday.
- 60% of Total Revenue Comes from Top 3 days



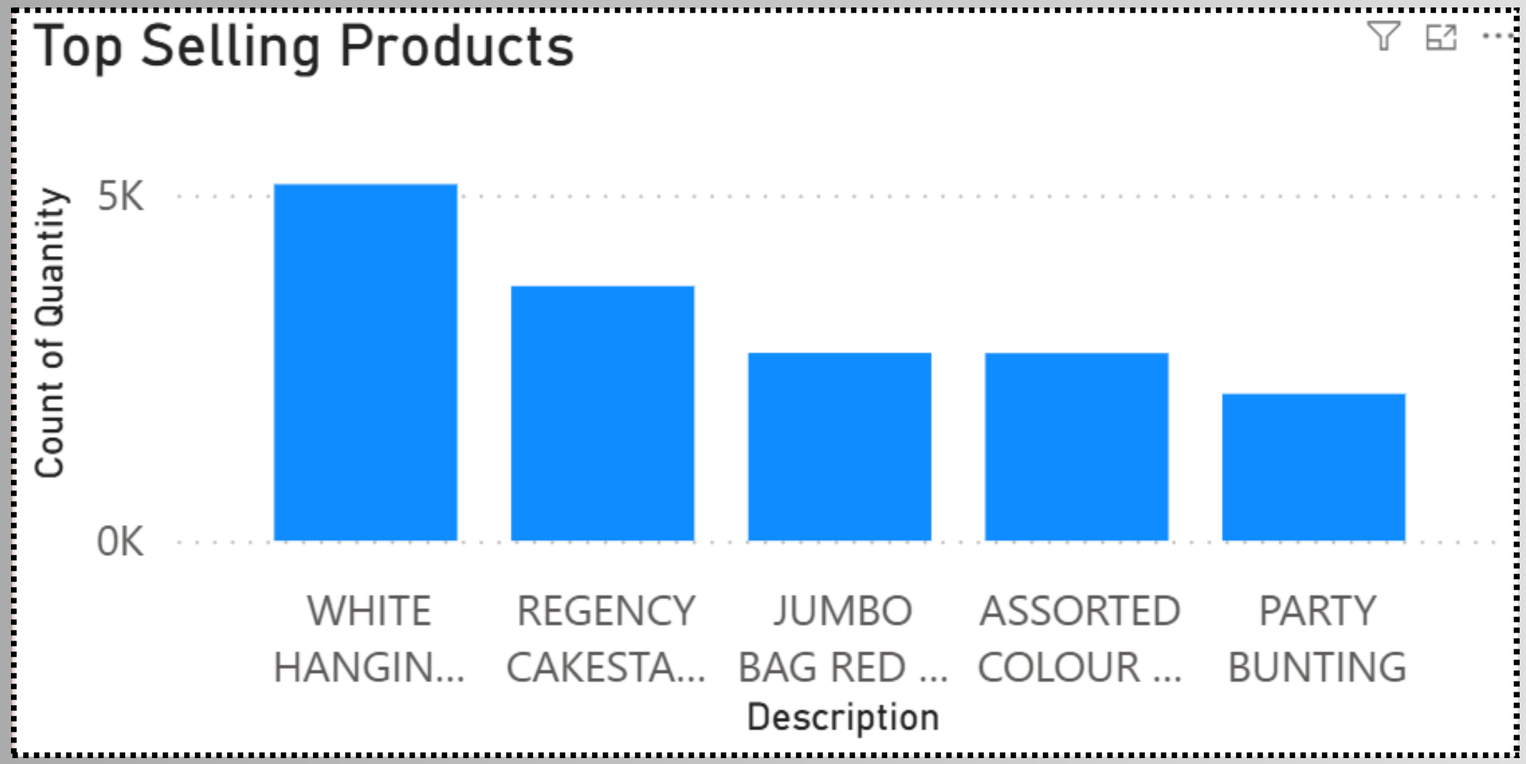
- Hourly sales peak around midday (10 AM – 1 PM)
- 50% of Total Revenue Comes from this window (10 AM - 1 PM)

2.Customer Insights



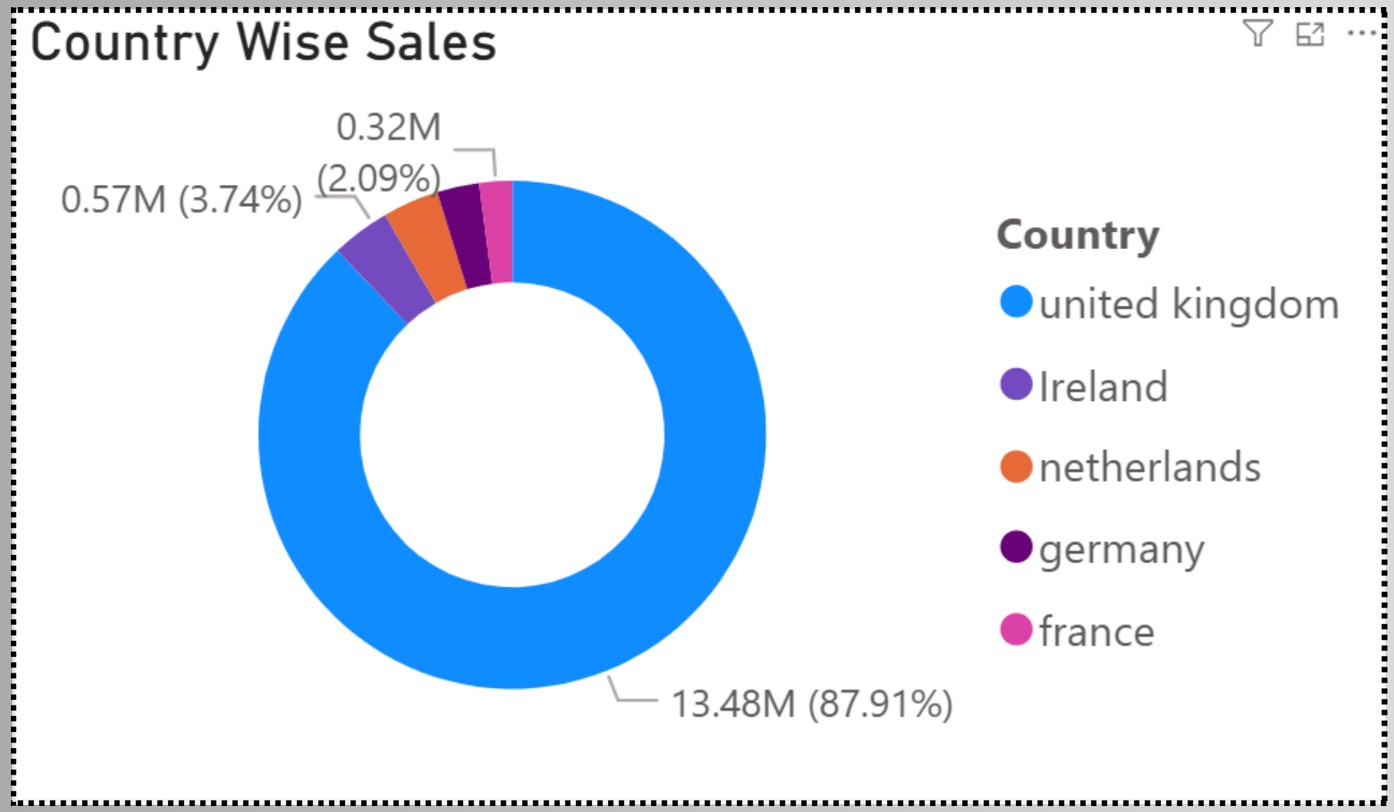
- A few customers contribute significantly higher revenue
- Indicates high dependency on top customers
- Top 5 customers contribute around 10% of Total revenue

3. Product Insights



- **Certain products (e.g., decorative and gift items) are high in demand**
- **Sales are concentrated among few top-selling items**

4. Country-wise Insights



- **United Kingdom dominates sales (~87% contribution)**
- **Other countries contribute very little → market concentration risk**

Conclusion

- **Sales are not evenly distributed across time, customers, or geography**
- **Business is heavily dependent on:**

1. Few customers
2. Few products
3. One major country

This creates both opportunities and risks

Recommendations

1. Business Actions

- Target low-performing days (weekends) with offers
- Expand marketing in other countries to reduce dependency on UK

2. Product Strategy

- Promote top-selling products more aggressively
- Bundle slow-moving products with bestsellers

3. Timing Optimization

- Run ads during peak hours (10 AM – 1 PM)
- Offer discounts during low activity hours

Executive Summary

This analysis of a 1M+ row e-commerce dataset reveals strong concentration patterns across time, geography, and customer segments.

- Seasonality: ~50% of total revenue is generated in just 4 months (Sep–Dec), with a peak in November
- Time Concentration: ~50% of sales occur within a 3-hour window (10 AM – 1 PM), indicating highly predictable buying behavior
- Geographic Dependency: ~87% of revenue comes from a single country (United Kingdom), creating market concentration risk
- Customer Contribution: Top 5 customers contribute ~10% of revenue, showing moderate concentration with growth potential

Analysis Steps

1.Data Cleaning

- Removed missing Values
- Duplicates column Removed
- Converted InvoiceDate to datetime format

2.Data Transformation

Created new columns:

- Month (from InvoiceDate)
- Weekday
- Hour

Calculated Column

- Total Sales = Quantity × Price

3.Aggregations

- Monthly sales trend
- Weekly sales trend
- Hourly sales trend